Progressive exercises

E1 · Direct application. Consider two events A and B such that $P(A) = 0{,}3$, $P_A(B) = 0{,}8$ and $P_{\overline{A}}(B) = 0.1$. Calculate $P(B)$, then $P_B(A)$.

E2 · Application and justification. The random variable X follows the binomial distribution with parameters 8 and $\frac{1}{4}$. Give the exact values of $P(X = 3)$, $E(X)$ and $V(X)$.

E3 · Method to organise. Let X a random variable taking integer values. For every integer k, set $F(k) = P(X \leq k)$. Express $P(X \geq 4)$ and $P(2 \leq X \leq 5)$ using F.

E4 · Choice of strategy. Five cards are drawn one after another, without replacement, from a 52-card pack containing four aces. Let X the number of aces obtained. The variable X follow a binomial distribution? Justify.

E5 · Synthesis or proof. We independently repeat a trial whose probability of success is $\frac{1}{2}$. Find the smallest number of trials for which the probability of obtaining at least one success is greater than or equal to 0.99.